Module 04: Cognition in Embodied Cognition — Thinking Through the Body

Learning Objectives

1. Define **embodied cognition** as thinking that is constitutively dependent on the body — not merely "thinking about" the body but thinking *through* bodily processes.
2. Analyze how **conceptual metaphors, image schemas, and body-based reasoning** rely on somatic generative models.
3. Apply the Active Inference framework to explain the **embodiment of abstract thought** — how even mathematical and logical reasoning has somatic underpinnings.

Introduction

The Cartesian tradition treats cognition as a disembodied process — a mind in a vat could think just as well as a mind in a body. Embodied cognition rejects this: the body is not merely a vehicle for the brain but a **constitutive element of cognition itself**. We understand "grasping" a concept through the same motor systems that grasp physical objects. We "weigh" arguments through the same interoceptive systems that assess physical weight. We "see" the point through the same visual predictive mechanisms that perceive physical scenes.

This module explores cognition as a body-based process — thinking that is shaped by, dependent on, and realized through somatic inference.

Key Concepts

1. Conceptual Metaphors and Body-Based Reasoning

Lakoff and Johnson's **conceptual metaphor theory** demonstrates that abstract thinking is systematically grounded in bodily experience:

- "Understanding is grasping" → The motor system's model of physical grasp is repurposed for intellectual apprehension

- "Importance is weight" → The proprioceptive model of heaviness is repurposed to assess significance
- "Time is space" → The spatial generative model developed for physical navigation is repurposed for temporal reasoning
- "Affection is warmth" → The interoceptive model of temperature is repurposed for social evaluation

In Active Inference terms, these metaphors work because the generative model's *structure* (the B matrix of spatial-temporal-causal relationships learned through bodily experience) is reused for abstract inference. The body provides the scaffolding for thought.

2. Image Schemas and Embodied Structure

Image schemas (Johnson, 1987) are pre-conceptual cognitive structures derived from bodily experience:

- **Container schema**: Inside/outside boundary (derived from the body as a container with skin as boundary) → Used for categorical reasoning ("in" the category, "outside" the set)
- **Source-Path-Goal schema**: Starting point, trajectory, destination (derived from bodily locomotion) → Used for reasoning about plans, arguments, and stories
- **Force schema**: Pushes, pulls, resistance (derived from physical interaction) → Used for reasoning about causation, obligation, and motivation
- **Balance schema**: Equilibrium, asymmetry (derived from vestibular experience) → Used for reasoning about justice, proportion, and aesthetics

These schemas are the **priors** of the embodied generative model — so deeply ingrained through universal bodily experience that they structure all subsequent cognitive development.

3. Gesture as Cognitive Process

Gesture is not merely expressive — it is **constitutive of thought**:

- People gesture when speaking on the phone (no visual audience) → gesture aids the speaker's own cognition

- Preventing gesture impairs problem-solving performance → gesture is part of the computational process
- Children gesture new concepts before they can verbalize them → the somatic generative model discovers structure before the verbal model can articulate it

In Active Inference, gesture is the motor generative model running its predictions in parallel with the verbal model — providing a spatial-dynamic simulation that supports abstract reasoning.

4. Extended Cognition and Cognitive Tools

The **extended mind hypothesis** (Clark & Chalmers, 1998) proposes that cognitive processes extend beyond the skull:

- A mathematician's scratchpad is part of their cognitive architecture — external memory that extends the generative model
- A surgeon's instrument is an extension of their proprioceptive model — the scalpel becomes part of the Markov blanket
- A musician's instrument is incorporated into the motor generative model — the piano keys become proprioceptive observations

The boundary of the cognitive agent is the **functional Markov blanket** — which may extend beyond the skin when tools are deeply integrated into the inference process.

Applications

- **Embodied mathematics education**: Teaching fractions through physical manipulation of fractional parts leverages the body-based generative model for quantity — children who physically divide and combine pieces develop more robust mathematical understanding than those learning purely symbolically.
- **Design thinking and prototyping**: Designers who build physical prototypes engage their somatic generative models — holding, turning, and manipulating a model engages proprioceptive and haptic inference that screen-based design cannot replicate. Embodied cognition explains why physical prototyping often yields insights that CAD modeling misses.

Conclusion

Thinking is not disembodied computation — it is inference through somatic generative models. Conceptual metaphors, image schemas, gesture, and extended cognition all demonstrate that the body is a constitutive element of cognition. The next module explores how embodied action — the complement of embodied cognition — completes the inference loop.